CS620 Capstone Project Fall ‘26

Family Finance Chat on GCP
From Chatbot to Interactive Avatar

Why this project is different: the system already has real users. This semester you will decide, on evidence, whether a student can practise a hard financial conversation by speaking to an avatar rather than typing to a chatbot — and then build it.

Project Overview

Family Finance Chat is a document-grounded AI system on Google Cloud Platform (GCP). Students learning to advise families about money practise against an AI family whose situation is drawn from the course corpus, and every question they ask is scored automatically for the instructor.

You will work across performance, deployment, evaluation, real-time media, and vendor assessment. The focus is practical engineering: measurable improvement and an honest recommendation.

What Was Built Last Semester

A prior CS620 team took the prototype into live use and hardened it. It is in weekly use, which is what makes the next step worth attempting.

- Ran a live beta for a student cohort on GCP infrastructure.
- Built observability: metrics, dashboards, and alerting on latency and usage.
- Established a reproducible evaluation of answer quality and grounding adherence.
- Assessed the OpenWebUI interface layer and kept it, with the reasoning documented.

What We Will Do This Semester

We will improve the system in two ways: make what exists fast and easy to change, then replace the chat box with a spoken, embodied client. The project is to evaluate the systems that make that leap possible, and implement the one that wins.

1) Make the current system performant and easy to update

- Bring the platform up to the current release, and make upgrades routine rather than exceptional.
- Move the instructor grading tool off a developer workstation onto the web, and automate deployment and testing.

2) Evaluate interactive avatar platforms

- Benchmark candidate real-time avatar services on latency, cost per student, and integration effort, and publish the numbers rather than impressions.
- Resolve consent, retention, and deletion for student voice and video before any session is recorded.

3) Implement the chosen system as a separate service

- Build the avatar as its own container beside the platform, never as a modification of it, so it can be removed without trace.
- Preserve the existing grounding and scoring path, so a spoken session is graded exactly like a typed one.

4) Prove it with students, and report honestly

- Run spoken sessions with a small cohort and compare question quality against typed sessions, using the scoring already in place.
- Write the finding up either way. A negative result is a result, and it is publishable.

What You Will Deliver

- A current, automatically deployed platform, with instructor grading available on the web.
- A written evaluation of avatar platforms: measured latency, cost, integration effort, and a recommendation.
- A working spoken-avatar client, running beside the platform and graded by the existing pipeline.
- Evidence comparing spoken and typed practice, with the method and data to reproduce it.

Project Sponsors

Mark Fedenia, Baird Professor of Finance, University of Wisconsin–Madison.
Role: sponsor and mentor, focused on real-user requirements, disciplined evaluation, and production-quality delivery.

Marshall Lund, Instructor, University of Wisconsin–Madison.
Role: Instructor for the students using the system. Domain expertise

Who Should Pick This Project

This capstone suits students who want to take a system real users depend on and make it do something it cannot do today.

- You want experience with real-time audio and video, not only request and response.
- You like evaluation: benchmark honestly, decide, and be willing to recommend no.
- You want cloud experience on GCP, including deployment and reliability.

Bottom line: you will make a live system faster and easier to change, and decide whether it should learn to talk.